

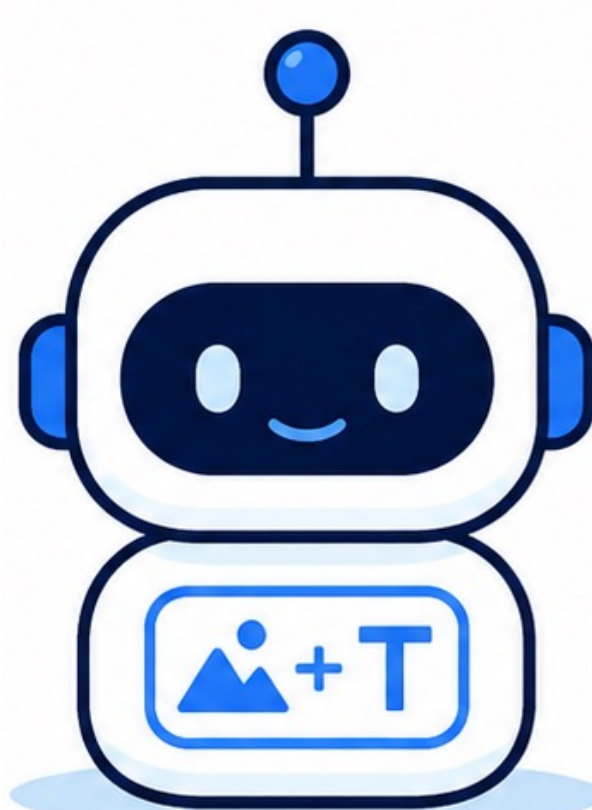
Input Problem

Which option from A-E should replace the question mark to follow the pattern present above the separating line?



Without TTS

Answer Directly

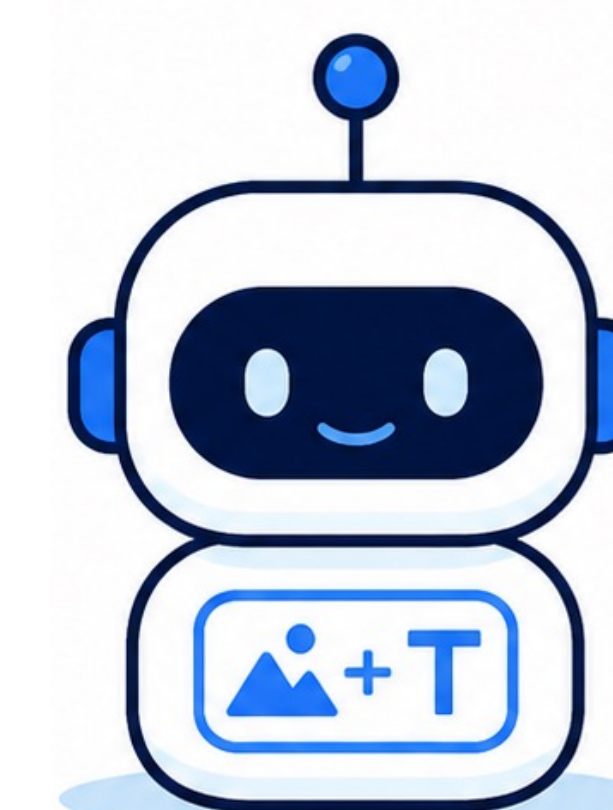


Answer: D

✗ Fails

Using TTS

Think step by step



Answer: B

✓ Correct

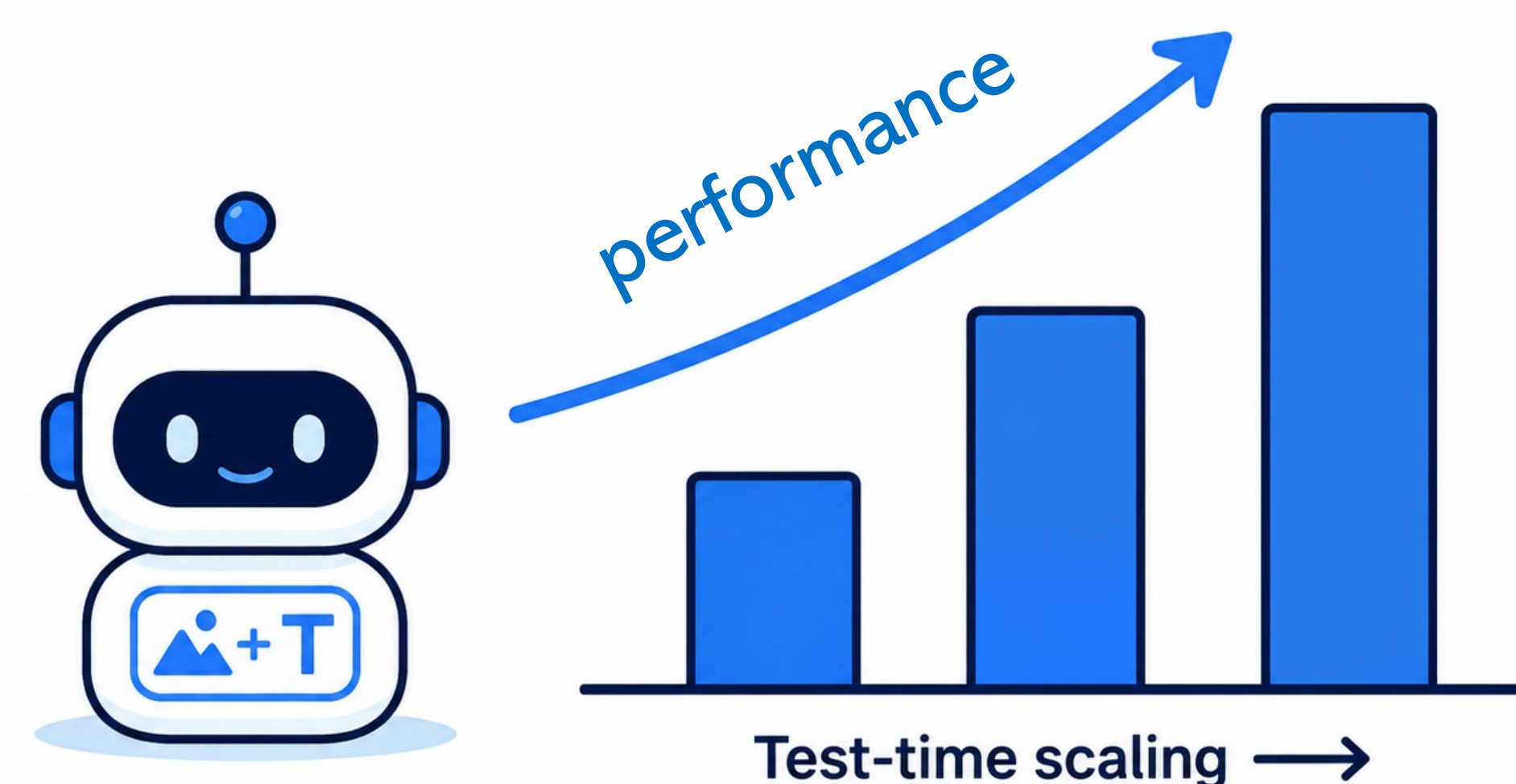
The shading alternates from dark background to white background.
The circle segment reduces by 45 degrees each time.
Therefore, the answer is B

Bigger isn't always better. Let *small VLMs* think



Finding 1

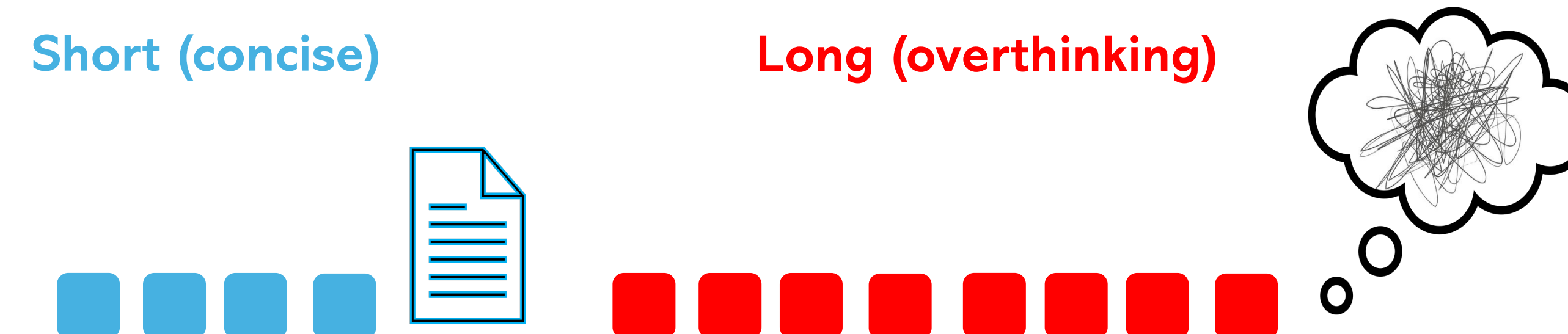
Small models benefit the most



- Small VLMs with TTS reach and even beat much larger VLMs.
- Save memory, compute and enhance speed!

Finding 2

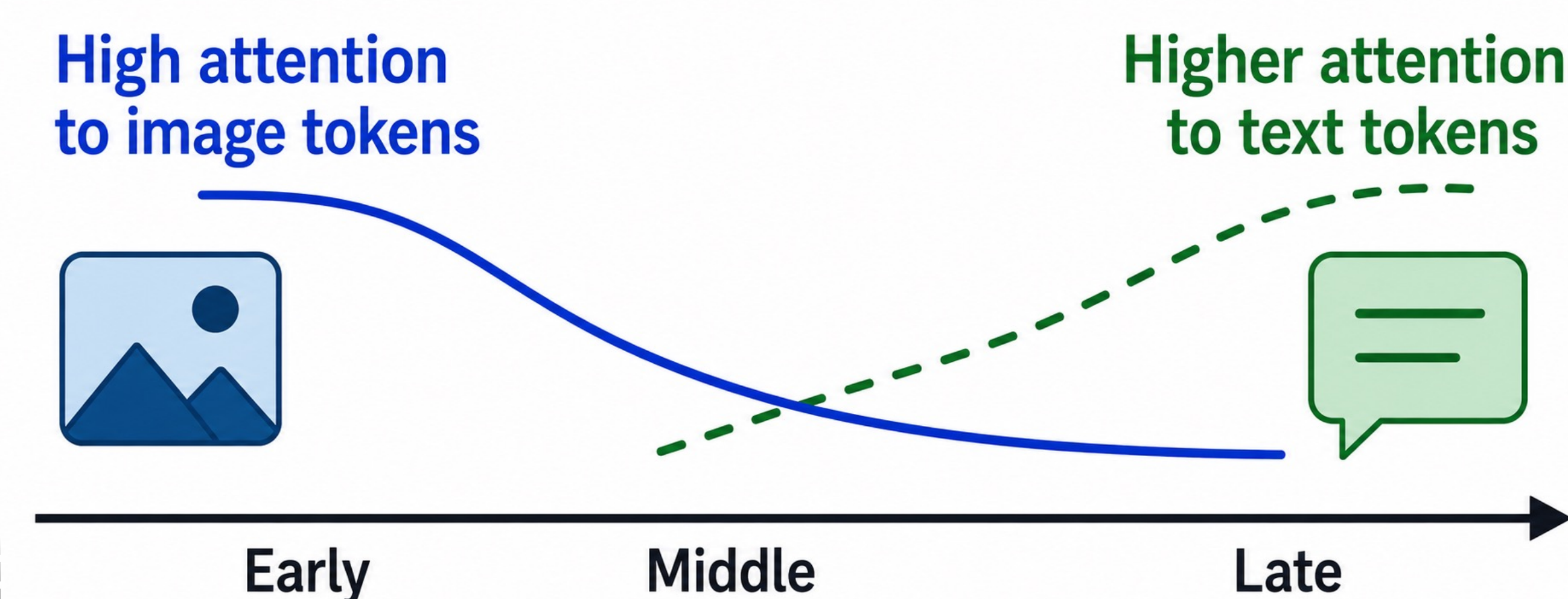
Too much compute can hurt performance on perceptual tasks



- On perception tasks, long generation makes VLMs lose focus, trigger hallucinations and worse answers
- Apply test-time scaling only when reasoning is needed

Finding 3

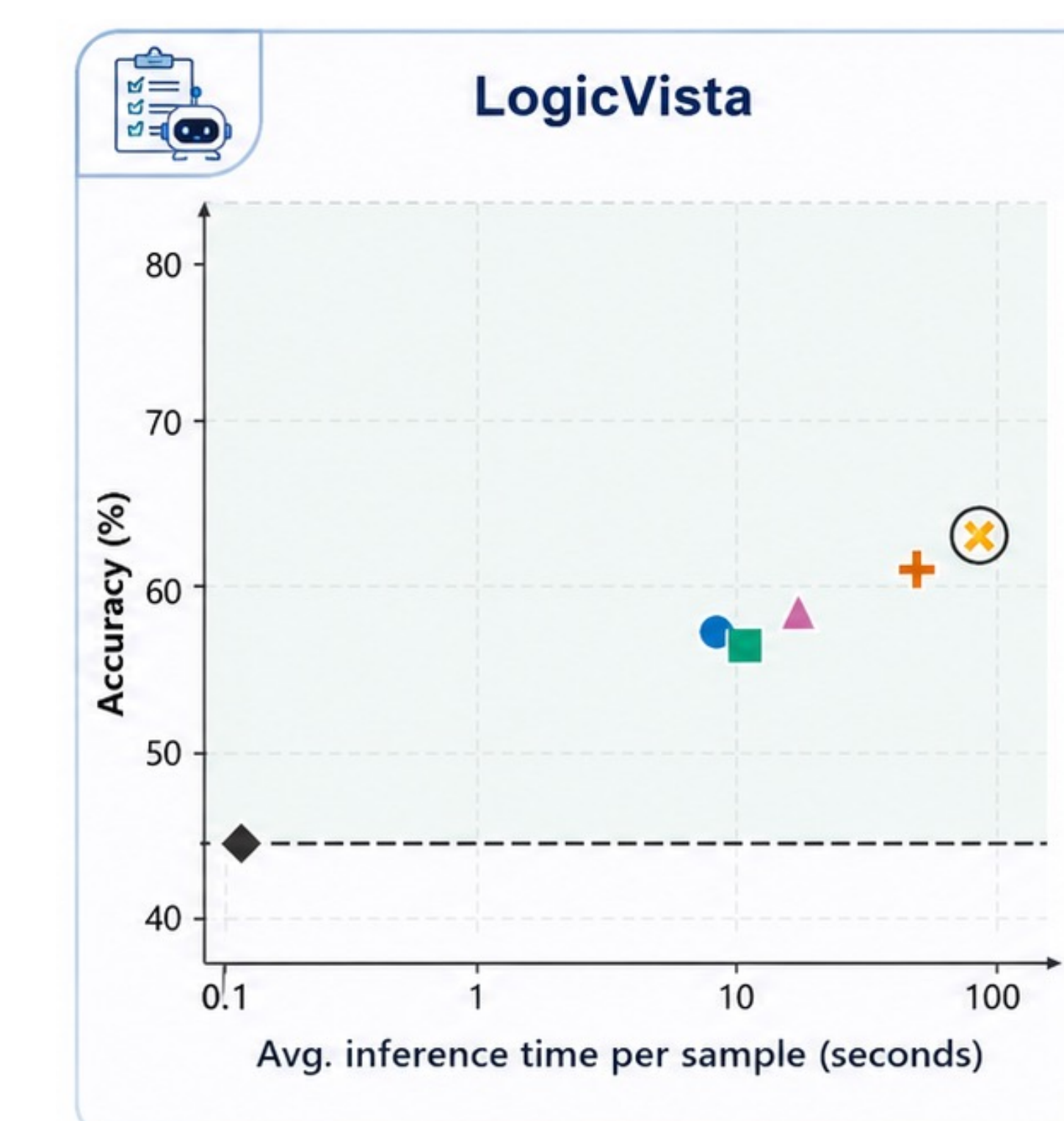
Reasoning relies on early visual encoding



- Later steps depend on visual information encoded early and absorbed into the text token representations
- Dropping image tokens later saves computation!

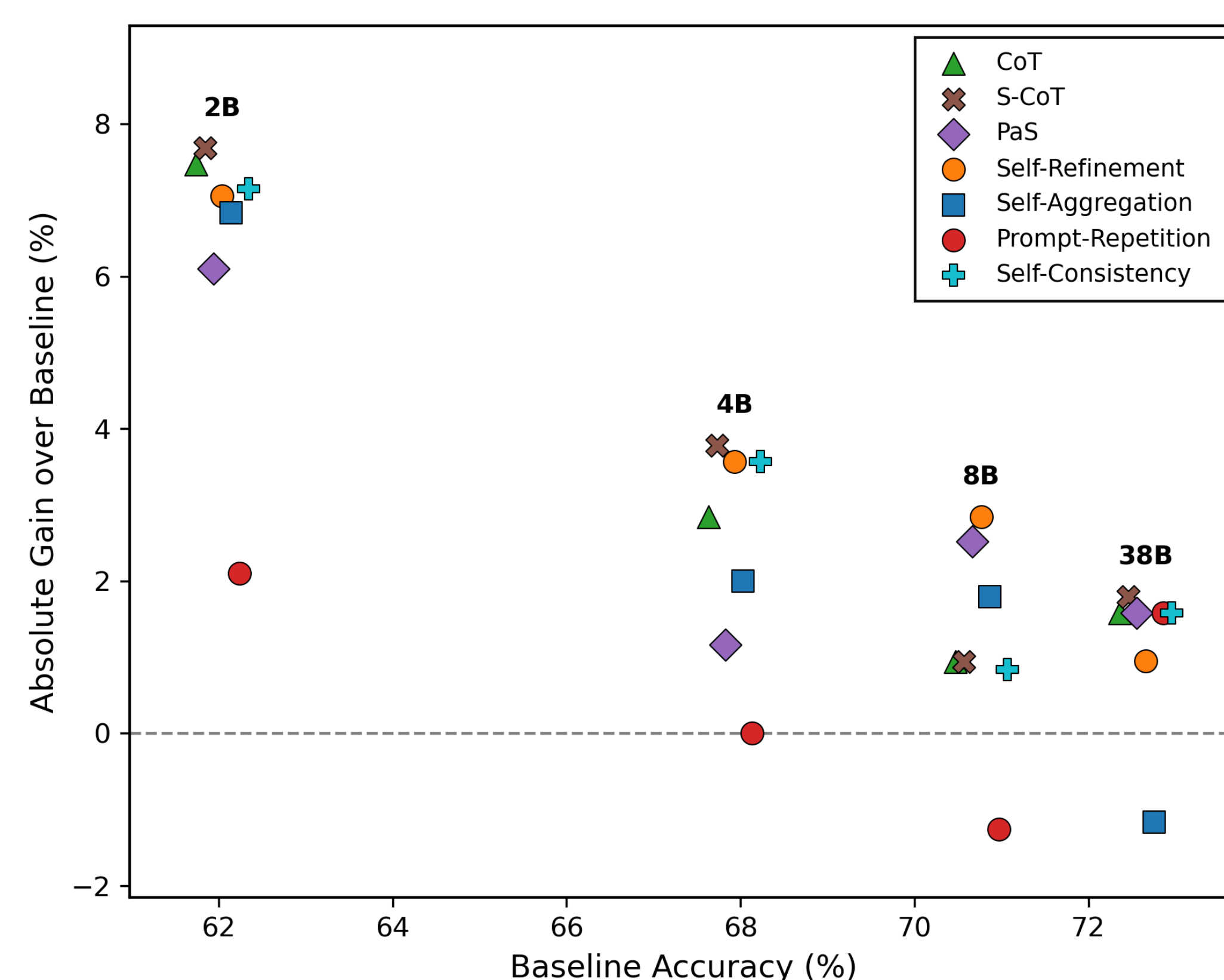
Finding 4

Pareto Frontier (compute vs accuracy)

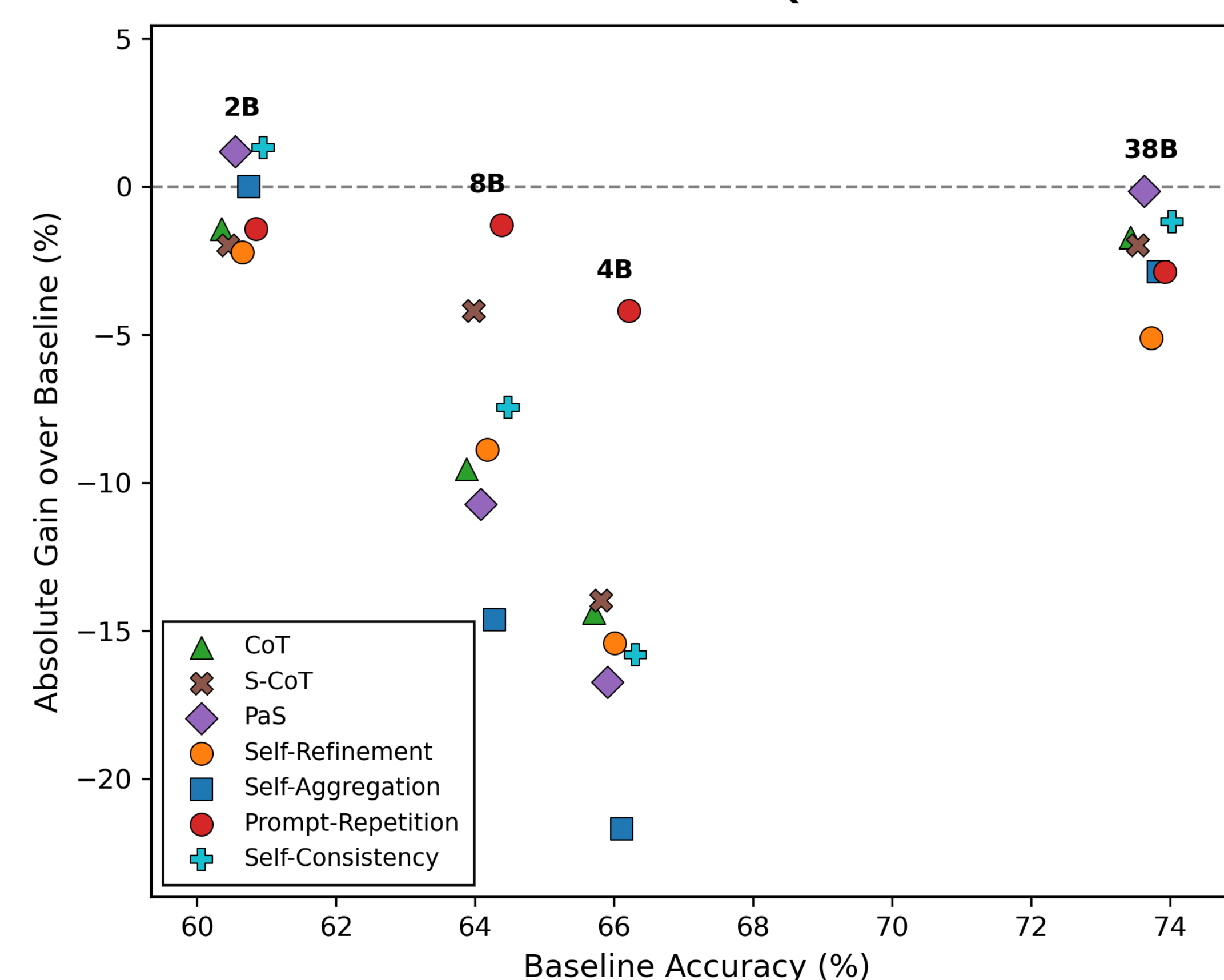


Absolute Gains (InternVL)

HallusionBench



RealWorldQA



TTS Performance (Qwen)

